

Hands-on Lab : INSERT, UPDATE, DELETE



Estimated time needed: 20 minutes

In this lab, you will learn some commonly used DML (Data Manipulation Language) statements of SQL other than SELECT. First, you will learn the INSERT statement, which is used to insert new rows into a table. Next, you will learn the UPDATE statement which is used to update the data in existing rows in the table. Lastly, you will learn the DELETE statement which is used to remove rows from a table.

Objectives

After completing this lab, you will be able to:

- Insert new rows into a table
- Update data in existing rows of the table
- Remove rows from a table

Concepts covered

How does the syntax of an INSERT statement look?

```
INSERT INTO table_name (column1, column2, ... )  
VALUES (value1, value2, ... )  
;
```

How does the syntax of an UPDATE statement look?

```
UPDATE table_name  
SET column1 = value1, column2 = value2, ...  
WHERE condition  
;
```

How does the syntax of a DELETE statement look?

```
DELETE FROM table_name  
WHERE condition  
;
```

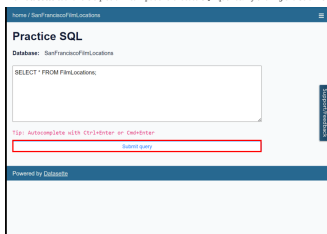
Introduction to Lab Environment

Software Used in this Lab

In this lab, you will use Datasette, an open source tool for exploring and publishing data. You can visit the datasette.io/lab repository here.

Working with Datasette

The Datasette tool offers a platform to input and execute SQL queries. By clicking the **Submit query** button, you can execute the SQL query.



Database Used in this Lab

The dataset used in this lab is an internal database.

Exploring the Database

Let us first explore the **Instructors** database using the **Datasette** tool:

1. If the first statement listed below is not already in the Datasette textbox on the right, then copy the code below by clicking on the little copy button on the bottom right of the codeblock below and then paste it into the textbox of the Datasette tool using either **Ctrl+V** or right-click in the text box and choose **Paste**.

```
SELECT * FROM Instructors;
```



2. Click **Submit Query**.
3. Now you can scroll down the table and explore all the columns and rows of the **Instructor** table to get an overall idea of the table contents.

ID	lastname	firstname	city	country
1	Cheng	Bob	Toronto	CA
2	Cheng	Bob	Toronto	CA
3	Voodman	Helen	Chicago	US

4. These are the column attribute descriptions from the **Instructor** table:

```
Instructor: 1  
(id, id |  
  unique identification number of the instructors,  
  lastname: last name of the instructors,  
  firstname: First name of the instructors,  
  city: name of the cities where instructors are located,  
  country: two-letter country code of the countries where instructors are located  
)
```

Exercise 1: INSERT

In this exercise, you will first go through some examples of using INSERT in queries and then solve some exercise problems by using it.

Task A

Example exercises on INSERT

Let us go through some examples of INSERT related queries:

1. In this example, suppose we want to insert a new single row into the **Instructor** table.

1. Problem:

Insert a new instructor record with id 4 for Sandip Suha who lives in Edmonston, CA into the "Instructor" table.

2. Solution:

```
INSERT INTO Instructors(id, lastname, firstname, city, country)  
VALUES(4, 'Suha', 'Sandip', 'Edmonston', 'CA');
```

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

4. Copy the code below by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

```
SELECT * FROM Instructors;
```

5. Your output resultset should look like the image below:



2. In this example, suppose we want to insert some new multiple rows into the **Instructor** table.

1. Problem:

Insert two new instructor records into the "Instructor" table. First record with id 5 for John Doe who lives in Sydney AU. Second record with id 6 for Jane Doe who lives in Dhaka, BD.

2. Solution:

```
INSERT INTO Instructors(id, lastname, firstname, city, country)  
VALUES(5, 'Doe', 'John', 'Sydney', 'AU'), (6, 'Doe', 'Jane', 'Dhaka', 'BD');
```

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.
4. Copy the code below by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

```
SELECT * FROM Instructor;
```



Task B

Practice exercises on INSERT

Now, let us practice creating and running some INSERT related queries.

1. Problem:
- Insert a new instructor record with id 7 for Antonio Ganguly who lives in Vancouver, CA into the "Instructor" table.

Hint

Follow example 1 of the INSERT exercise.

```
INSERT INTO table_name(column1[,column2], [column3], [column4], [column5])
VALUES('value1', 'value2', 'value3', 'value4', 'value5');
SELECT * FROM table_name;
```

Solution

```
INSERT INTO Instructor(id, lastname, firstname, city, country)
VALUES(7, 'Ganguly', 'Antonio', 'Vancouver', 'CA');
SELECT * FROM Instructor;
```

Output

```
INSERT INTO Instructor(id, lastname, firstname, city, country)
VALUES(7, 'Ganguly', 'Antonio', 'Vancouver', 'CA');
SELECT * FROM Instructor;
```

Results

9 commands ran successfully

SELECT * FROM Instructor

id	lastname	firstname	city	country
1	Aliya	Ravi	Toronto	CA
2	Chang	Paul	Toronto	CA
3	Choudhary	Prerna	Chicago	US
4	Datta	Sandip	Toronto	CA
5	Doe	John	Dublin	AE
6	Elia	John	Osaka	JP
7	Ganguly	Antonio	Vancouver	CA
8	Ryan	Steve	Burling	GB
9	Samant	Ramesh	Hyderabad	IN

2. Problem:
- Insert two new instructor records into the "Instructor" table. First record with id 8 for Steve Ryan who lives in Burling, GB. Second record with id 9 for Ramesh Samant who lives in Hyderabad, IN.

Hint

Follow example 2 of the INSERT exercise.

```
INSERT INTO table_name(column1[,column2], [column3], [column4], [column5])
VALUES('value1', 'value2', 'value3', 'value4', 'value5');
SELECT * FROM table_name;
```

Solution

```
INSERT INTO Instructor(id, lastname, firstname, city, country)
VALUES(8, 'Ryan', 'Steve', 'Burling', 'GB'), (9, 'Samant', 'Ramesh', 'Hyderabad', 'IN');
SELECT * FROM Instructor;
```

Output

```
INSERT INTO Instructor(id, lastname, firstname, city, country)
VALUES(8, 'Ryan', 'Steve', 'Burling', 'GB'), (9, 'Samant', 'Ramesh', 'Hyderabad', 'IN');
SELECT * FROM Instructor;
```

Results

9 commands ran successfully

SELECT * FROM Instructor

id	lastname	firstname	city	country
1	Aliya	Ravi	Toronto	CA
2	Chang	Paul	Toronto	CA
3	Choudhary	Prerna	Chicago	US
4	Datta	Sandip	Toronto	CA
5	Doe	John	Dublin	AE
6	Elia	John	Osaka	JP
7	Ganguly	Antonio	Vancouver	CA
8	Ryan	Steve	Burling	GB
9	Samant	Ramesh	Hyderabad	IN

Exercise 2: UPDATE

In this exercise, you will first go through some examples of using UPDATE in queries and then solve some exercise problems by using it.

Task A

Example exercises on UPDATE

Let us go through some examples of UPDATE related queries:

1. In this example, we want to update one column of an existing row of the table.

1. Problem:

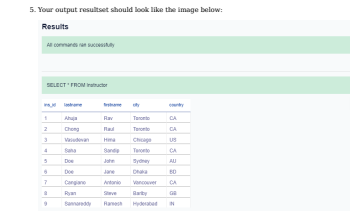
Update the city for Sandip to Toronto.

Solution:

```
UPDATE Instructor
SET city='Toronto'
WHERE lastname='Sandip';
```

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.
4. Copy the code below by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

```
SELECT * FROM Instructor;
```



2. In this example, we want to update multiple columns of an existing row of the table.

1. Problem:

Update the city and country for Doe with id 5 to Dubai and AE respectively.

Solution:

```
UPDATE Instructor
SET city='Dubai', country='AE'
WHERE id=5;
```

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.
4. Copy the code below by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasette tool. Then click **Submit query**.

```
SELECT * FROM Instructor;
```

5. Your output resultset should look like the image below:

Practice SQL

Database: instructors

1 SELECT * FROM instructors;

Tip: Autocomplete with Ctrl+Space or Cmd+Space

Submit query

Results

All commands ran successfully

SELECT * FROM instructors

id	id	first_name	last_name	email	country
1	1	John	Smith	john_s@instructor.com	CA
2	2	Neena	Kharwal	neena_k@instructor.com	CA
3	3	Lex	DeHaan	lex_d@instructor.com	CA
4	4	Alexander	Toh	alex_t@instructor.com	CA
5	5	Baer	Patel	baer_p@instructor.com	IN
6	6	Chen	Adam	chen_a@instructor.com	CA
7	7	Beaumont	Chen	beaumont_c@instructor.com	CA
8	8	Scott	Stevens	scott_s@instructor.com	CA
9	9	Deena	Dei	deena_d@instructor.com	CA

Task B

Practice exercises on UPDATE

Now, let us practice creating and running some UPDATE related queries.

1. Problem:
- Update the city of the instructor record to Markham whose id is 1.

Hint

Follow example 1 of the UPDATE exercise.

UPDATE table_name
SET (column1[, column2[, ...]])
WHERE (search_condition);
SELECT * FROM table_name;

Solution

UPDATE instructors
SET city = 'Markham'
WHERE id = 1;
SELECT * FROM instructors;

Output

1 Update successful

Tip: Autocomplete with Ctrl+Space or Cmd+Space

Submit query

Results

All commands ran successfully

UPDATE instructors

id	id	first_name	last_name	email	country
1	1	John	Smith	john_s@instructor.com	CA
2	2	Neena	Kharwal	neena_k@instructor.com	CA
3	3	Lex	DeHaan	lex_d@instructor.com	CA
4	4	Alexander	Toh	alex_t@instructor.com	CA
5	5	Baer	Patel	baer_p@instructor.com	IN
6	6	Chen	Adam	chen_a@instructor.com	CA
7	7	Beaumont	Chen	beaumont_c@instructor.com	CA
8	8	Scott	Stevens	scott_s@instructor.com	CA
9	9	Deena	Dei	deena_d@instructor.com	CA

2. Problem:
- Update the city and country for Scudip with id 4 to Dhaka and BD respectively.

Hint

Follow example 2 of the UPDATE exercise.

UPDATE table_name
SET (column1[, column2[, ...]])
WHERE (search_condition);
SELECT * FROM table_name;

Solution

UPDATE instructors
SET city = 'Dhaka', country = 'BD'
WHERE id = 4;
SELECT * FROM instructors;

Output

1 Update successful

Tip: Autocomplete with Ctrl+Space or Cmd+Space

Submit query

Results

All commands ran successfully

UPDATE instructors

id	id	first_name	last_name	email	country
1	1	John	Smith	john_s@instructor.com	CA
2	2	Neena	Kharwal	neena_k@instructor.com	CA
3	3	Lex	DeHaan	lex_d@instructor.com	CA
4	4	Alexander	Toh	alex_t@instructor.com	BD
5	5	Baer	Patel	baer_p@instructor.com	IN
6	6	Chen	Adam	chen_a@instructor.com	CA
7	7	Beaumont	Chen	beaumont_c@instructor.com	CA
8	8	Scott	Stevens	scott_s@instructor.com	CA
9	9	Deena	Dei	deena_d@instructor.com	CA

Exercise 3: DELETE

In this exercise, you will first go through an example of using DELETE in a query and then solve an exercise problem by using it.

Task A

Example exercise on DELETE

Let us go through an example of a DELETE related query:

1. In this example, we want to remove a row from the table.
1. Problem:
- Remove the instructor record of Doe whose id is 6.
2. Solution:
- DELETE FROM instructors
WHERE id = 6;

3. Copy the solution code above by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of Custom SQL query of the Datasense tool. Then click Submit query.
4. Copy the code below by clicking on the little copy button on the bottom right of the codeblock below and paste it to the textbox of the Datasense tool. Then click Submit query.
- DELETE FROM instructors;

5. Your output resultset should look like the image below:

Practice SQL

Database: instructors

1 DELETE FROM instructors;

Tip: Autocomplete with Ctrl+Space or Cmd+Space

Submit query

Results

All commands ran successfully

DELETE FROM instructors

id	id	first_name	last_name	email	country
1	1	John	Smith	john_s@instructor.com	CA
2	2	Neena	Kharwal	neena_k@instructor.com	CA
3	3	Lex	DeHaan	lex_d@instructor.com	CA
4	4	Alexander	Toh	alex_t@instructor.com	CA
5	5	Baer	Patel	baer_p@instructor.com	IN
6	6	Chen	Adam	chen_a@instructor.com	CA
7	7	Beaumont	Chen	beaumont_c@instructor.com	CA
8	8	Scott	Stevens	scott_s@instructor.com	CA
9	9	Deena	Dei	deena_d@instructor.com	CA

Task B

Practice exercise on DELETE

Now, let us practice creating and running a DELETE related query:

1. Problem:
- Remove the instructor record of Hima.

Hint

Follow example 1 of the DELETE exercise.

DELETE FROM table_name
WHERE (column_name) = 'value';
SELECT * FROM table_name;

Solution

DELETE FROM instructors
WHERE first_name = 'Hima';
SELECT * FROM instructors;

• Output

```
1 SELECT * FROM employees;
2 INSERT INTO employees (first_name, last_name, email, phone_number, hire_date, job_id, salary, commission_pct)
3 VALUES ('Neena', 'Kharwal', 'NEENA.K@oracle.com', 9901, TO_DATE('2015-09-17', 'YYYY-MM-DD'), 'ANALYST', 3000, 0);
4 UPDATE employees SET salary = salary * 1.1;
5 DELETE FROM employees WHERE last_name = 'Neena';
```

Results

1 row(s) returned

1 row(s) returned

1 row(s) returned

id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct
1	Neena	Kharwal	NEENA.K@oracle.com	9901	2015-09-17	ANALYST	3000	0

Thank you for completing the Hands-on Lab : INSERT, UPDATE, DELETE! where you learnt to perform operations on tables like inserting and removing rows, and updating the data in existing rows.

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